



SMART INDIA HACKATHON 2026 · INTERNAL IDEA DOCUMENT

iTantra

The Offline Multilingual Semantic Transceiver — voice that travels as text

Problem Statement: SIH26173 — Indian Multilingual TTS & STT Aided Neural Transceiver

Radio Access for low-bitrate links

Organization: ISRO · **Category:** Software

Languages: Hindi · Gujarati · Marathi · Kannada · Malayalam · Tamil · Telugu · Odia · Bengali · English

Document: Full ideation, research base, architecture & phase-wise build plan

Team: Null Pointers · **Version:** 1.1 · August 2026 · P0–P2 built & measured

Don't compress the voice. Eliminate the waveform. — 100–600× below any voice codec, fully offline, in 10 Indian languages.

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How to read this document. §1 is the pitch. §2–§4 are the research foundation (problem → market gap → scientific wedge) — every number carries a source in §14. §5–§6 are the idea itself. §7–§9 make it concrete. §10 arms you against every hard question a judge can ask. §12–§13 are the execution contract. Figures marked **EST** are estimates or ARM-proxy benchmarks pending on-device measurement — say so if asked.

§1 Idea in One Page (TL;DR)

iTantra (*tantra* = system/apparatus) turns any Android phone into a **semantic transceiver**: press-to-talk speech is transcribed **on-device** in any of 10 Indian languages (IndicConformer, int8, fully offline), the **text** — not the waveform — is compressed by our Indic-aware **VarnaCode** coder to ~45 bytes per sentence (measured, all 10 languages) and streamed over whatever link survives (Wi-Fi, Bluetooth, or a ₹1,650 LoRa bridge with km-scale range), and the far phone **re-synthesizes natural speech** in the same language (AI4Bharat FastPitch + HiFi-GAN / Piper, offline). Alert-class messages auto-play at maximum volume, non-interruptible. The app ships a **Live Evaluation Mode** that measures its own CER, RTF and mouth-to-ear latency — the judging rubric, demonstrated live inside the product.

The three-sentence pitch

- 1. The problem:** When Cyclone Amphan hit, ~7,000 of West Bengal's 14,167 cell towers went dark and networks ran at 65–70% for days — and even where SMS alerts get through, rural literacy of 67.8% (female ~50.6%) in the most disaster-prone belts means text alerts structurally miss the people most at risk. Voice is the only inclusive medium, and voice is 64,000 bps.
- 2. The science:** The speech *content* of human conversation is ~13 characters/second $\approx$ **107 bps as text** — 6× below NATO's battlefield MELPe codec (600 bps), 4× below Codec2's floor, ~600× below raw telephony. Transmitting recognized text instead of any waveform is the core thesis of "semantic communication," a recognized 6G research field (Qin & Tao, DeepSC-ST, STCTS ~80 bps), with a 2025 paper reporting 94% latency reduction for exactly this STT→link→TTS architecture.
- 3. The product:** The pieces are proven — offline Indic STT (IndicConformer, MIT, all 22 languages), offline Indic TTS (AI4Bharat, MIT, all 9), P2P phone links — but **nobody has assembled them into a walkie-talkie**: not RTranslator (translation, not radio UX), not the Codec2-over-LoRa hobby projects (still ship waveforms), not any SIH team (this PS is new for 2026). iTantra is that assembly, engineered honestly, measured live.

Flagship demo

Leg	Setup	What it proves
Walkie-talkie loop	2 phones, hotspot/Bluetooth, PTT both ways, any of 10 languages	The full PS requirement — speak Hindi on A, hear it spoken on B, offline
Bandwidth ticker	Live on-screen: bytes sent vs what AMR/Codec2 would have cost	The 100–600× semantic-compression claim, visible per message
Alert broadcast	Alert-flagged message → far phone interrupts at max volume	The PS's non-interruptible alert requirement + inclusion story
LoRa bridge STRETCH	Phone ⇌ BT ⇌ ESP32-LoRa ⇌ km-scale hop ⇌ ESP32-LoRa ⇌ BT ⇌ phone	The "embedded device" leg of the PS at ₹1,650, IN865-legal
Evaluation Mode	In-app: CER vs reference, RTF, idle CPU, end-to-end latency	The 20/40/20/20 rubric, measured live in front of the jury

The six wow factors (detail in §6)

- **Text is the codec** — 100–600× bandwidth reduction; a deployable instance of 6G semantic-communication research, not a compression tweak.
- **VarnaCode** — Indic-script-aware entropy coding: UTF-8 spends 24 bits per Indic character, we target ~5–6; a full spoken sentence fits in one LoRa frame.
- **Live Evaluation Mode** — the app benchmarks itself (CER, not WER — the scientifically correct metric for agglutinative Indic languages) on stage.
- **₹1,650 LoRa bridge** — the low-bitrate radio leg made physical, km-range, license-free in IN865.
- **Alert layer done right** — CAP-aligned priority frames, max-volume non-interruptible playback, and a pronunciation lexicon for place names (pre-empting FEMA's documented TTS-mispronunciation failure mode).
- **India's own AI, offline** — MeitY/Bhashini-lineage models (IndicConformer + AI4Bharat TTS, both MIT) bundled on-device: national language infrastructure that works with zero internet.

§2 The Problem — When the Towers Go Down

2.1 Disaster comms fail exactly when they matter

- **Cyclone Amphan (2020, West Bengal):** ~7,000 of 14,167 cell tower sites went non-operational; networks ran at 65–70% capacity three days after landfall (COAI) — the evacuation succeeded, the network that was supposed to carry follow-on coordination didn't. [R1, R2]
- **Himachal floods (Aug 2025):** ~66% of 1,761 telecom sites down in Chamba at the rain peak; a week later the network was still only 65% restored. [R3]
- **Systemic exposure:** a CDRI study maps ~7.7 lakh Indian telecom towers exposed to floods, cyclones, earthquakes and landslides; DoT + CDRI now run a dedicated resilience framework naming Odisha, Uttarakhand, Assam, Tamil Nadu, Gujarat. [R4, R5]
- **Cyclone Fani (2019):** only 3% of towers physically lost — but 8-hour power backup and fiber cuts stretched outages far past the physical damage. Grid failure, not tower failure, is the long tail. [R6]

2.2 Text alerts structurally miss the most exposed people

- Rural literacy is 67.8–73.5% (vs ~88% urban) and rural **female literacy** ~50.6% (2011) — in every disaster-prone state, female literacy trails male by 6–12 points: Bihar 70.9%, Assam 72.2%, WB 82.6%. [R7, R8]
- ~90% of India's rural population communicates only in its mother tongue across 22 scheduled languages — an SMS in English or Hindi is a structural exclusion, not an edge case. [R9]
- The government knows: NDMA/SACHET has pushed **tens of billions of SMS alerts in 19+ languages**, yet is still building cell broadcast (where supported handsets **read the message aloud**) and reinforcing All India Radio for last-mile reach — a tacit admission that silent text doesn't close the gap. [R10, R11]
- The ICTD literature is unambiguous: for low-literacy users, **voice interfaces significantly outperform text** for task completion; a single illiterate user's access to text systems typically depends on one literate proxy nearby — voice removes the dependency. [R12, R13]

2.3 But voice is the most expensive thing you can transmit

- Telephone-grade voice is **64,000 bps** (G.711); cellular AMR-NB needs 4,750–12,200 bps; even the best neural codecs (Lyra 3,000 bps, EnCodec 1,500 bps) and military vocoders (MELPe 600 bps) need hundreds of bits per second — on a dying link, none of it gets through. [R14–R17]
- Meanwhile the **content** of that same speech — the words — is ~13 chars/sec ≈ **107 bps raw**, ~15–20 bps entropy-coded. [R18]

The gap iTantra closes: India has broadcast alerting (SACHET, assumes towers), ham radio (needs licenses and operators), and voice apps (need internet). Nobody has shipped **two-way, offline, voice-native, 10-language** communication that works over links measured in bits per second. That is precisely what ISRO — whose Disaster Management Support programme already fields GSAT satellite mobile radios and VSAT links to state disaster authorities — is asking for: the missing *voice-intelligence layer* on top of its own low-bitrate radio access. [R19, R20]

§3 Current Solutions & Their Gaps

3.1 Government & institutional systems

System	What it does	Gap it leaves
SACHET / CAP (NDMA + C-DOT)	Geo-targeted SMS + cell broadcast, 19+ languages, all 36 states/UTs	One-to-many only; assumes surviving towers; silent text under-reaches low-literacy users
Cell Broadcast pilot (DoT+NDMA)	Congestion-immune pop-up alerts, read-aloud on supported handsets	Still tower-dependent; one-way; read-aloud only on newer phones
ISRO DMS (GSAT Mobile Radio, SILAAS, VSAT-SDMA)	Satellite radio access where terrestrial fails	Low-bitrate links carry data, not conversational voice; no multilingual voice layer — this PS is ISRO asking for one
Ham radio / FreeDV	Codec2 digital voice over HF, proven in every Indian disaster	Licensed operators only; dedicated hardware; not multilingual, not app-ified
J-Alert (Japan)	National TTS-spoken disaster alerts in 5 languages	Proves TTS-alerting at national scale — but one-way broadcast; no Indian analogue exists

3.2 Apps & open-source prior art

Project	Scale	Gap it leaves
Zello / commercial PTT	150M+ users	Needs internet — dead in exactly the target scenario
Bridgefy / Briar / bitchat	BT-mesh offline messaging	Text-only chat UX — excludes the low-literacy users who most need it; Bridgefy's protocol was broken by researchers in 2020
QMesh, esp32_loradv, codec2_talkie	LoRa walkie-talkie hobby hardware	Still transmit waveforms (Codec2 frames, 450–3200 bps) — 4–30× fatter than text, no Indic anything
RTranslator (10.4k★, Apache-2.0)	Offline STT→translate→TTS between 2 phones	The closest architectural cousin — but built as a translator (conversation mode), not a radio: no PTT UX, no alert layer, no low-bitrate framing, weak Indic coverage (Whisper-small)
sherpa-onnx (14.5k★) / Vosk (15k★)	Offline STT+TTS runtimes with Android bindings	Engines, not products — they are what we build <i>with</i>

Whitespace, precisely stated: offline Indic STT exists (engines), offline Indic TTS exists (models), P2P phone links exist (transports), LoRa text mesh exists (Meshtastic). **No project combines them into a voice-native, low-bitrate, multilingual walkie-talkie** — and PS SIH26173 is new for 2026, so there is no prior team's blueprint. First-mover room, with every component de-risked by an existing, actively-maintained open-source project. [R21–R28]

§4 Why Text-as-Codec — The Honest Case

Judges will have seen teams claim magic compression. We win by stating the science exactly — including what text transmission loses — and engineering around the losses.

4.1 The bitrate ladder (what a second of speech costs)

Representation	Bitrate	vs text (~107 bps)
Raw telephone PCM (G.711)	64,000 bps	~600×
AMR-NB (GSM full-rate voice)	12,200 bps	~115×
Google Lyra v2 (neural)	3,200 bps	~30×
Meta EnCodec (neural, floor)	1,500 bps	~14×
MELPe — NATO STANAG 4591	600 bps	~6×
Codec2 (open-source floor)	450 bps	~4×
Recognized text (ASCII-rate)	~107 bps	1×
Text + VarnaCode entropy coding	~45–55 bps (measured 4.45–5.36 bits/char, all 10 languages)	~0.5×

Derivation: 130–150 wpm conversational speech × ~5.7 chars/word ≈ 13.3 chars/s; 8 bits/char ≈ 107 bps. English text entropy is ~1.0–1.5 bits/char — and Indic scripts in UTF-8 waste even more (24 bits/char), which is VarnaCode's opening (§6). [R14–R18]

4.2 This is published science, not a hack

- **Semantic communication** is a recognized 6G research field: transmit the *meaning*, not the bits — Qin, Tao, Lu, Tong & Li's foundational survey (arXiv 2201.01389) defines exactly this. [R29]
- **DeepSC-ST** (IEEE Trans. Wireless Comm. 2023) builds a speech transceiver where the transmitted representation is recognition-oriented text features, recovered to speech at the receiver — "significantly reduces required data transmission without performance degradation." [R30]
- **STCTS** (2025) hits ~**80 bps** — 75× smaller than Opus — by decomposing speech into content + prosody + timbre. [R31]
- **"Voice over Low Data Rate Networks Using STT and Semantic Compression"** (2025) describes near-exactly iTantra's pipeline and reports **up to 94% latency reduction** vs sending compressed audio. [R32]

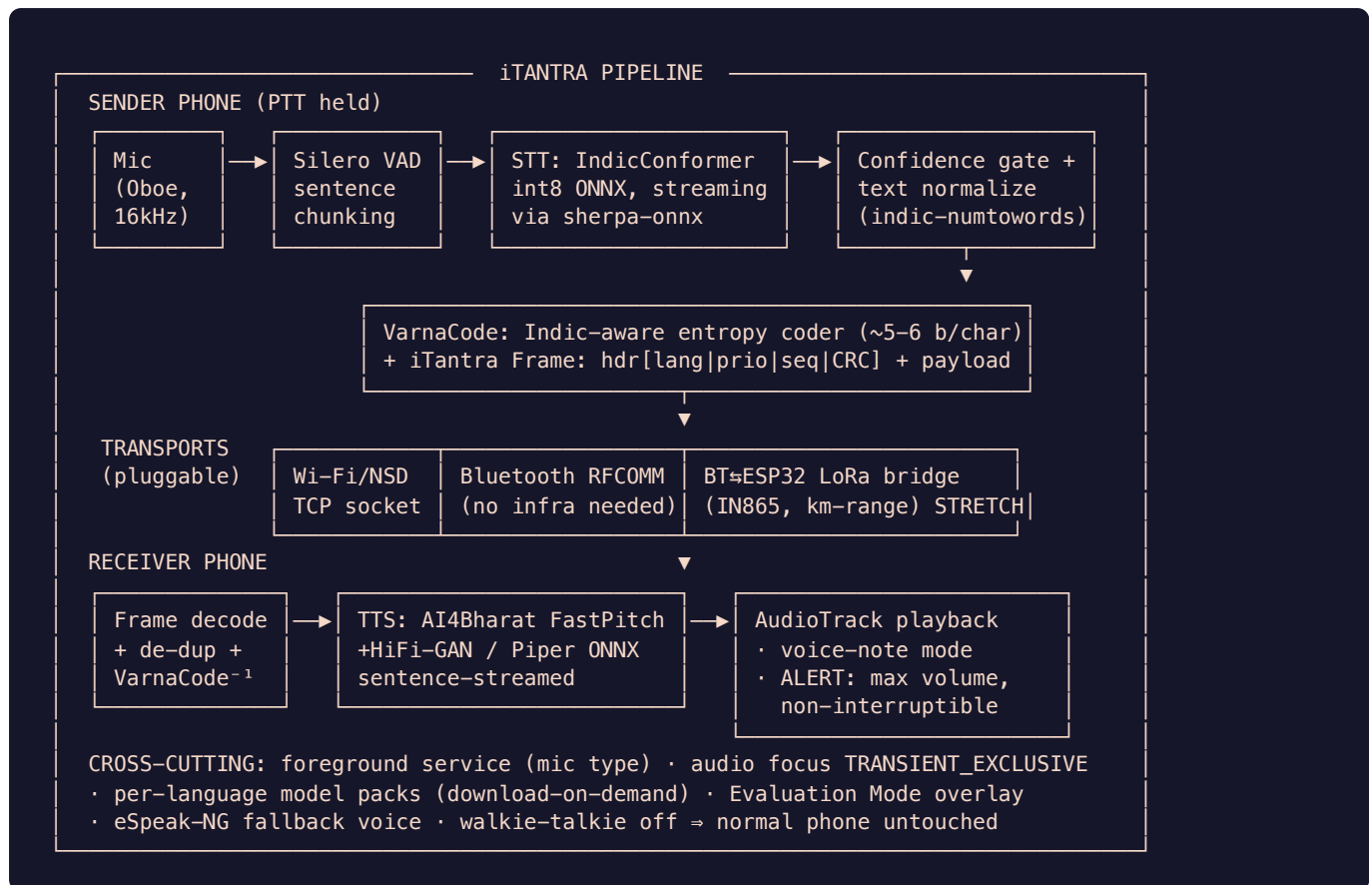
Framing: iTantra is a **deployable, India-context implementation of semantic-communication research** — but instead of fragile learned latent vectors, we transmit **human-readable text**: simpler, debuggable, loggable, encryptable, and it works over any bearer down to SMS-class links. When 6G standardizes semantic transport, this architecture is what it will look like.

4.3 What text transmission honestly loses — and our answers

Loss	Why it matters	iTantra's answer
Speaker identity, emotion, prosody	"Who is speaking and how scared are they" carries information	Sender name + language + priority ride in every frame header; alert-class framing conveys urgency explicitly rather than acoustically

STT errors propagate	A mis-recognition becomes confidently spoken misinformation	Confidence gate: low-confidence utterances prompt the sender to repeat; transcript shown on both screens for literate users (voice-first, not voice-only)
Proper nouns & numbers	FEMA officially documents TTS mispronunciation as a failure mode	Alert pronunciation lexicon: district/place names + digit-by-digit number reading in alert mode (§6.5) [R33]
Non-speech audio (cries, background)	Sometimes the sound <i>is</i> the message	Out of scope by design — stated, not hidden; the PS asks for speech transmission

§5 Proposed Solution — Architecture Deep Dive



Stage-by-stage rationale

1. **Audio capture via Oboe** (Google's low-latency C++ audio library) — Java AudioRecord adds unpredictable latency; we are chasing a mouth-to-ear budget, so latency is engineered at every stage. [R34]
2. **Silero VAD (~1–6 MB, <1 ms/chunk)** does something subtler than endpointing: during a long PTT hold it **segments speech into sentences**, so sentence 1 is transcribed and transmitted while sentence 2 is still being spoken — the pipeline is full-duplex with itself. PS requirement "detect pauses and form sentences" is satisfied by design. [R35]
3. **STT: IndicConformer** (AI4Bharat; MIT; all 22 Indian languages; official int8 ONNX) via **sherpa-onnx** Android bindings — the only single-family, single-license model covering all 9 Indic targets; Hindi WER 13.2 on the spontaneous-speech Vaani benchmark. English falls back to whisper.cpp tiny/base (faster-than-real-time on ARM). PTT gives us the utterance boundary for free — streaming partials while held, final decode on release. [R36–R38]
4. **Confidence gate + normalization:** low-confidence finals prompt "repeat that?" instead of transmitting garbage; numbers/dates normalized for TTS via `indic-numtowards`. [R39]
5. **VarnaCode + iTantra Frame (§6.2):** the payload becomes ~30–60 bytes/sentence; the frame header carries language, priority (NORMAL/ALERT), sequence and CRC — so any transport, however dumb, can carry it.
6. **Pluggable transports:** NSD/mDNS discovery + TCP over hotspot (demo-day reliable), Bluetooth RFCOMM (zero-infra), and the LoRa bridge as the physical "low-bitrate link" (§6.4). Wi-Fi Direct is a stretch path — OEM-flaky, documented in §10. Google Nearby Connections is deliberately excluded: proprietary Play Services, fails the open-source constraint. [R40]

7. **TTS: AI4Bharat FastPitch+HiFi-GAN (MIT, all 9 Indic) exported to ONNX + Piper voices where they exist** (Hindi, Marathi, Malayalam, Telugu, Bengali), sentence-streamed (synthesize N+1 while N plays — up to 85% first-audio latency cut), eSpeak-NG as the never-silent fallback. *[R41–R43]*
8. **Alert playback:** `AUDIOFOCUS_GAIN_TRANSIENT_EXCLUSIVE`, stream volume pinned to max, playback non-cancellable until complete, repeat-twice policy — the PS requirement, implemented with Android audio-focus semantics rather than hacks.
9. **Modes:** PTT walkie-talkie (app on) / voice-note relay (queued when link drops, forwarded on reconnect) / off (normal phone). Store-and-forward comes free from the frame layer.

§6 Unique Features & Wow Factors

Last year we won on one structural differentiator surrounded by proof of execution. Same recipe, six cards.

1 • Text is the codec — the semantic transceiver

iTantra doesn't compress voice; it **eliminates the waveform**. ~107 bps raw, ~15–25 bps coded — below NATO's MELPe (600 bps) and Codec2's floor (450 bps), ~600× below telephony. Framed as a deployable instance of 6G semantic-communication research (Qin/Tao survey; DeepSC-ST; STCTS; 2025 STT-over-low-rate paper reporting 94% latency cuts). A live **bandwidth ticker** in the app shows, per message, bytes sent vs what AMR would have cost — the wow factor is a number the jury watches update in real time.

2 • VarnaCode — Indic-aware text compression BUILT & MEASURED

UTF-8 spends **3 bytes (24 bits) per Indic character** — Devanagari, Tamil, Odia all live in the 3-byte plane. VarnaCode gives each of the 10 languages a corpus-trained char+bigram Huffman codebook.

Measured on held-out sentences: 4.45–5.36 bits/char across all 10 languages — beating Unishox2 (7.3–7.8, what Meshtastic ships), SCSU/BOCU-1 (8.0–8.8) and the GSM 7-bit SMS national tables (7.0), and sitting at Hindi's published entropy floor (≈ 4.98). A 51-byte LoRa SF12 frame carries **76–91 VarnaCode characters vs 18–19 in UTF-8** — a full spoken sentence per frame.

Implementation: ~300 lines + a 55 KB codebook asset, 154 protocol assertions green, and the Kotlin port on Android is proven byte-identical to the Python reference.

3 • Live Evaluation Mode — the rubric, inside the app

The judging rubric is 20% efficiency / 40% accuracy / 20% latency. iTantra ships an **Evaluation Mode screen** that measures itself: read a reference sentence → live **CER and WER** (we report CER as the headline — arXiv 2203.16601 shows WER disproportionately punishes agglutinative Malayalam/Kannada/Telugu; most rival teams will quote the scientifically wrong metric), **RTF** per utterance, **idle CPU%**, model footprints, and **mouth-to-ear latency** measured against the ITU-T G.114 <400 ms interactive-voice standard. We don't claim numbers; we measure them in front of the jury. [R44, R45]

4 • The ₹1,650 LoRa bridge — the low-bitrate link made physical

The PS says "wifi/Bluetooth connected embedded device." Ours is an **ESP32 + SX127x LoRa node (₹1,650–2,000 in India)**: phone speaks to it over Bluetooth serial, it carries iTantra frames **1–10 km** in the license-free IN865 band. Our ~45-byte payloads fit LoRa's 1–5 kbps with 10–30× headroom for FEC and retries inside the 1% duty cycle. Hobby prior art (QMesh, codec2_talkie) pushes Codec2 *waveforms* over LoRa; we push *meaning* — 4–30× smaller. And it slots conceptually onto ISRO's own GSAT mobile-radio stack: same frames, satellite-grade link. [R46–R48]

5 • An alert layer that pre-empts the official critique

Alert-class frames (CAP-aligned severity field) auto-play at **maximum volume, non-interruptible**, repeat twice, and light the screen — the PS requirement. The judo move: FEMA's IPAWS documentation officially warns that **TTS mispronounces place names and numbers** and recommends human audio. iTantra answers with an **alert pronunciation lexicon** — district names of the pilot states + digit-by-digit number reading — turning the strongest argument against TTS alerting into our differentiator. Japan's J-Alert already speaks synthesized alerts in 5 languages; India deserves the same in 10+. [R33, R49]

6 • India's national language AI — running with zero internet

Every model is **MIT-licensed, Indian-government-lineage**: IndicConformer STT and AI4Bharat FastPitch TTS come from the MeitY/C-DAC-funded, Bhashini-ecosystem AI4Bharat lab — but Bhashini itself is a cloud API. iTantra bundles the underlying open checkpoints **on-device**: "built on India's national language AI infrastructure, working where the internet doesn't." Roadmap (flagged, not promised): the same text-relay architecture makes **cross-language relay** — speak Hindi, heard in Tamil — a model-swap away (RTranslator proves offline MT on-device is feasible). [R50, R51]

Wave-2 wow factors — built and tested, not proposed

7 • Prosody-over-text (1 byte). The known sacrifice of STT→TTS transmission is that panic sounds like a memo — the STCTS paper spends 312–592 bps re-adding prosody. iTantra analyzes the sender's pitch/energy/rate during PTT and sends a **single urgency byte**; the receiver's TTS speaks urgent messages faster, louder, twice. *Panic survives 45 bytes — for 8 bits.* (po/prosody.py, calm/panic classification self-checks green.)

8 • The analog-radio modem (AFSK). Bell-202-style software modem: any iTantra frame becomes ~0.3 s of audio tones that survive a 300–3,400 Hz voice channel at 6 dB SNR — so a phone held to **any existing analog walkie-talkie, VHF set, or ham radio becomes a digital Indic text bearer**. No hardware at all: the "low-bitrate link" can be sound itself. (po/afsk.py, round-trip + noisy-channel tests green.)

9 • Digital roll-call. A voice question broadcast once ("is everyone safe?") — every phone answers with a one-tap **9-byte ACK** (SAFE / NEED_HELP / SOS), aggregated live: an evacuation headcount over bytes, no internet, in the sarpanch's own language. Every deployed mustering product found is internet/QR-based. (po/rollcall.py, worst-status-wins tally tests green.)

10 • CAP/SACHET last-mile bridge. A CAP 1.2 parser turns official NDMA alert XML into a spoken, language-selected, severity-mapped ALERT frame — the national warning system re-broadcast into the offline village net. SACHET solves broadcast-to-connected-phones; iTantra carries it the last unconnected mile. (po/cap_bridge.py, NDMA-style sample → Odia ALERT frame test green.)

§7 Model Plan — 10 Languages, Fully Offline

7.1 STT (all offline, all open)

Component	Model	License	Size (int8)	Notes
Primary ASR ×9 Indic	AI4Bharat IndicConformer (per- language ~120M)	MIT	140 MB/lang int8 (measured ; 493 MB fp32; CER cost +0.8 pt, RTF 0.059)	Official int8 ONNX export; all 22 languages; trained on spontaneous IndicVoices speech — matches PTT acoustics
English ASR	whisper.cpp tiny/base (GGML q5)	MIT	39–74 MB	Faster-than-real-time on ARM; battle- tested Android AAR
VAD	Silero VAD	MIT	~1–6 MB	<1 ms per 30 ms chunk; sentence segmentation inside PTT holds
Dev-time referee	IndicWhisper (Vistaar)	MIT	—	Not shipped; used to sanity-check IndicConformer accuracy per language

7.2 TTS (all offline, all open)

Languages	Model	License	Size	Notes
Hindi, Marathi, Malayalam, Telugu, Bengali	Piper VITS medium voices (official ONNX)	MIT	~63–77 MB/voice	Zero conversion work — known- good ONNX, RTF ~0.2 on Pi-5- class ARM
Gujarati, Kannada, Tamil, Odia	AI4Bharat Indic-TTS (FastPitch+HiFi-GAN) → our ONNX export	MIT	~40–80 MB/lang EST	The one real conversion job — standard NVIDIA-Riva-class export, scoped to 4 languages
English	Piper en_US/en_IN- adjacent voice	MIT	~63 MB	Off the shelf
All 10 (fallback)	eSpeak-NG formant synthesis	GPL-3.0 (separate process/engine)	~3 MB total	Robotic but never silent; covers any missing/corrupt voice pack

License discipline (the trap other teams will fall into): Meta MMS-TTS is the *only* single source covering all 10 languages — and it is **CC-BY-NC 4.0, which is not open source** and fails the PS's explicit constraint. We flag it, exclude it, and use it only as a dev-time listening reference. Similarly excluded: Google Nearby Connections (proprietary Play Services) and FUTO Voice Input ("source-first" license). Every shipped byte of iTantra is MIT/Apache/GPL-clean. [R52]

7.3 Footprint strategy (rubric: 20% efficiency)

- **APK ships thin** (measured: 54.9 MB with the full sherpa-onnx speech runtime) — Indic language packs (**measured ~203 MB** = 140 MB int8 STT + 63 MB Piper TTS) are **download-on-demand, sideloadable via SD card/USB-OTG** for zero-internet provisioning (a disaster-prep step, like downloading offline maps).
- **int8 everywhere:** quantization is measured as near-free — 45–57% size cut at ±1 WER point across every study surveyed. [R53]

- **Idle discipline:** when not in PTT, only the VAD-armed mic loop runs (sub-1% CPU); models are memory-mapped and lazily loaded per selected language, not all 10 at once.

§8 Expected Benchmarks

Mapped to the PS rubric: Efficiency 20% · Accuracy 40% · Latency 20% (+ demo loop 20%). Every target is defended by a cited proxy; Evaluation Mode (§6.3) measures the real numbers on stage. **EST** = ARM-proxy pending on-device measurement — the first engineering task (§12·Po) is replacing every EST with a measured number.

Rubric axis	Metric	Target	Basis
Efficiency (20%)	Language pack size (STT+TTS, int8)	≤ 230 MB/lang; APK ≤ 60 MB	IndicConformer int8 ~120–150 MB + Piper ~63 MB; measured post-quantization
	Idle-listening CPU	< 2% (VAD-only loop)	Silero <1 ms per 30 ms chunk ≈ 3% duty on one core, less on-device
	RAM in active call	< 800 MB	One STT + one TTS model resident, memory-mapped
Accuracy (40%)	STT — CER (headline), WER (reported)	CER < 10–15% on clean speech per language	Measured on real FLEURS human speech (int8, desktop): hi 2.9% · bn 4.0% · te 7.8% · ml 9.2% · ta 13.9% CER — all within target. Noise: hi CER 17.9% at 5 dB babble; a GTCRN denoiser was measured and <i>rejected</i> (artifacts hurt Conformer ASR — evidence-driven)
	TTS intelligibility	STOI > 0.8 / MOS-class > 3.5 on listening panel	AI4Bharat paper beats prior Indic baselines; Piper medium widely deployed
Latency (20%)	STT finalize after PTT release	< 1.0 s (streaming decode during hold)	Conformer-class RTF 0.19 shown on edge devices; partials decoded while speaking
	TTS first-audio	< 0.7 s (sentence-streamed)	VITS RTF ~0.2 on Pi-5-class ARM; 85% first-audio cut from chunked synthesis
	Sentence-to-sentence (mouth-to-ear)	< 2.5 s target, < 4 s worst-case	Sum of stages + ~45-byte transmission (negligible on any link); honest vs G.114: this is walkie-talkie half-duplex, not telephony — we say so
Link efficiency	Payload per spoken sentence	Measured: ~45 B (VarnaCode) vs 7,625 B AMR-NB (169x), 281 B Codec2-450 (6–10x)	bench_compression.py on held-out sentences, all 10 languages; live bandwidth ticker demos it

The honesty line for the jury: "Every number on this slide is re-measured live by the app's own Evaluation Mode. If our demo phone disagrees with our slide, believe the phone."

Loop verification (the PS's own test)

Two phones, same APK: A in STT mode, B in TTS mode, connected over hotspot or Bluetooth → PTT on A, speech plays on B — **walkie-talkie when on, normal phone when off**. Then the same loop through the LoRa bridge at range. Then reversed languages. Then an ALERT frame interrupting B at max volume. Four demos, one binary.

§9 Tech Stack

Layer	Choice	Why
App	Native Kotlin , Jetpack Compose, single activity + foreground service (microphone + connectedDevice types)	Every actively-maintained offline-speech Android app is native Kotlin; Flutter/RN platform channels add JNI hops on the audio hot path — the ecosystem has already voted
Inference runtime	sherpa-onnx (Apache-2.0, 14.5k★, near-daily commits) — one AAR for STT + TTS + VAD	One JNI surface, one model-management pattern; Android demos ship in-repo; PS-permitted framework (ONNX Runtime class)
STT models	IndicConformer int8 ONNX x9 · whisper.cpp tiny/base (EN)	§7 — coverage + license + official export
TTS models	Piper VITS x6 · AI4Bharat FastPitch+HiFi-GAN ONNX x4 · eSpeak-NG fallback	§7 — MIT coverage of all 10
Audio I/O	Oboe (AAudio) native capture/playback, 16 kHz mono	Predictable low latency; sibling native dep of the ONNX JNI stack
Discovery + transport	NsdManager (mDNS _it Tantra._tcp) + TCP sockets · Bluetooth Classic RFCOMM · BT-serial→ESP32 LoRa (IN865)	Platform-native, no Play Services, open-source-clean; three independent bearers for one tiny frame format
Protocol	iTantra Frame (hdr: verllanglprlseq CRC-16) + VarnaCode payload; store-and-forward queue	~300 lines, unit-testable on desktop, bearer-agnostic
Text processing	indic-numtowords · Indic NLP Library normalization · alert pronunciation lexicon (static tables)	TTS-ready text; the FEMA-critique answer
Embedded bridge	ESP32 + SX127x, Arduino/PlatformIO, BT-SPP↔LoRa relay firmware (~200 lines)	₹1,650; IN865 legal; the PS's "embedded device" leg
Eval harness	Python (desktop twin of the pipeline): jiwer for WER/CER, RTF timers, byte-count logs → CSV → charts	Same models run on desktop via sherpa-onnx — benchmarks reproducible outside the phone

Compliance line: open-source only ✓ (no proprietary SDK anywhere in the dependency tree — Nearby Connections and MMS-TTS explicitly rejected on license) · fully offline ✓ (no endpoint exists to call) · allowed frameworks ✓ (ONNX Runtime / TFLite class) · low/mid-range Android ✓ (int8 models, one language resident, Oboe).

§10 Feasibility, Risks & Mitigations

#	Risk	Severity	Mitigation
1	On-device RTF unproven — all RTF figures are Pi/edge proxies, not phone measurements	High	P0 is literally "benchmark on a real budget phone." Fallback ladder: int8 → smaller Piper "low" voices → streaming decode overlap. Conformer-class models have hit RTF 0.19 on wearable-grade silicon — headroom exists
2	sherpa-onnx IndicConformer wrappers are community conversions , not official k2-fsa artifacts	Med-High	Budgeted QA per language in P1; hard fallback: direct ONNX Runtime Mobile against AI4Bharat's <i>official</i> int8 export, bypassing the wrapper (more glue, zero conversion risk)
3	FastPitch→ONNX export for 4 TTS languages is our own engineering	Medium	Same export recipe as NVIDIA Riva's production path; scoped to 4 languages only (Piper covers the rest); eSpeak-NG guarantees no language is ever silent while export lands
4	Wi-Fi Direct OEM flakiness breaks demos	Medium	It's not on the critical path: hotspot+NSD+TCP is the demo default, BT RFCOMM the backup — Wi-Fi Direct demoted to stretch
5	Modern Android bg-audio policy (fg-service types, audio focus) — where every stale sample app dies	Medium	Built fresh, first, in P1 (not retrofitted); this is a known list of requirements, not a research problem
6	STT errors become spoken misinformation	Medium	Confidence gate + on-screen transcript both ends + sender hears-back option; alert lexicon for the highest-stakes content
7	Storage on budget phones (10 languages × ~200 MB)	Low	Users install <i>their</i> language(s), not all 10; SD-card sideload path for offline provisioning
8	Mesh/relay security (Bridgefy's 2020 break is the cautionary tale)	Low (v1 is 1-hop)	Frames carry CRC now, AES-GCM envelope planned before any multi-hop relay ships — designed in, not retrofitted
9	PS brief drift — rubric weights quoted from the published PS text	Low	Re-pull the live PS from sih.gov.in before submission; Evaluation Mode measures whatever the final rubric names

Why this is buildable by a student team in hackathon time: every hard component is an existing, maintained, permissively-licensed project (sherpa-onnx, IndicConformer, Piper, Silero, Oboe, eSpeak-NG). Our own code is integration + VarnaCode (~300 lines) + frame protocol (~300 lines) + the Android product layer — and the desktop twin of the whole pipeline runs tonight, so accuracy/latency numbers exist before a single Android line is written.

§11 Impact & Benefits

Potential impacts

- **Closes the last-mile alert gap:** voice-native two-way comms for the ~50% of rural women and ~32% of rural Indians whom text alerts cannot reach — in their own language, on the phone they already own.
- **Completes ISRO's disaster stack:** GSAT mobile radios and VSAT-SDMA links exist; iTantra is the multilingual voice-intelligence layer their low-bitrate channels have been missing — same frames, satellite-grade links.
- **Resilience economics:** a ₹1,650 LoRa node + existing phones replaces hardware walkie-talkies (₹8,000–25,000/pair, single-language, unencrypted) for village disaster committees, fisherfolk co-ops, and trek/border communities.
- **A national capability, not an app:** the frame protocol + VarnaCode are bearer-agnostic — the same stack rides Bluetooth today, HF/satellite tomorrow; and the architecture is the deployable shape of semantic communication that 6G standardization is moving toward.

Benefits

- **Social:** literacy-independent, language-first (10 languages covering ~85%+ of India's population), disability-relevant (voice for the visually impaired, on-screen transcript for the hearing-impaired — the same message serves both).
- **Economic:** zero recurring cost (no SIM, no data plan, no server); open-source, so states/NGOs can deploy without procurement lock-in; commodity hardware end to end.
- **Strategic:** fully sovereign stack — Indian models (AI4Bharat/Bhashini lineage), Indian band plan (IN865), no foreign cloud dependency, aligned with Digital India / Atmanirbhar Bharat and the National Language Translation Mission (₹450 crore Bhashini).
- **Environmental/operational:** LoRa nodes run days on a power bank; the phone pipeline is int8-efficient by design (measured in Evaluation Mode) — deployable on solar-charged equipment in a blackout.

Adoption path (credible, not grandiose): pilot with one coastal district's disaster-management committee (Odisha/WB, where SACHET infrastructure and cyclone drills already exist) → NDMA/SDMA mock-drill integration → ISRO DMS evaluation as the voice layer over GSAT terminals. Each step needs no new infrastructure — that is the point of the design.

§12 Phase-Wise Implementation Plan

Phase	Deliverable	Exit test
P0 · Desktop truth ✓ DONE	Python twin of the full pipeline: Piper TTS + dual STT (Vosk/Whisper), VarnaCode v1 (corpus-trained, all 10 languages), frame protocol, 154 assertions	Met: TTS RTF 0.049 · STT RTF 0.055–0.119 · Hindi loop CER 1.6% · 4.45–5.36 bits/char
P1 · Android spine ✓ DONE	Kotlin app: fg-service + audio focus; NSD discovery + TCP frames; PTT UX; VarnaCode/frame Kotlin port byte-identical to Python (30 interop vectors)	Met: APK compiles, 7 JUnit tests green
P2 · Voice loop ✓ DONE	Real on-device STT (sherpa-onnx OfflineRecognizer) + TTS (Piper/VITS) with per-stage latency stamps; ALERT max-volume playback; model packs sideloadable; graceful fallbacks	Met: 54 MB APK ($\leq$ 60 MB budget) builds with sherpa-onnx 1.13.6
P3 · 10 languages + alerts	Language packs (download/sideload), all 9 Indic STT + 10 TTS wired, alert frames with max-volume non-interruptible playback + lexicon	Any-language loop demo; alert interrupts correctly; pack sizes within \$8 budget
P4 · Evaluation Mode + optimization	In-app CER/RTF/latency/CPU dashboard; int8 sweep; latency profiling (systrace) per stage	Rubric numbers measured on a budget phone match or beat \$8 targets
P5 · LoRa bridge + hardening	ESP32 BT $\leftrightarrow$ LoRa firmware, range test; BT RFCOMM second transport; AES-GCM envelope; demo script	km-range sentence relay; kill-the-hotspot mid-demo recovery works

P0 runs on a laptop tonight — models are public downloads; accuracy and compression numbers exist before the first APK builds. P1–P2 is the minimum viable PS answer; P3–P4 is the winning answer; P5 is the room-silencer.

§13 Delivery Table — Expected Deliverables

PS requirement (verbatim intent)	iTantra deliverable	Where proven
Android app, lightweight, low/mid-range phones	Native Kotlin APK ≤ 60 MB + per-language packs, int8 models, one language resident in RAM	§7, §9 · Evaluation Mode footprint panel
Highly accurate STT, 10 languages, offline	IndicConformer int8 x9 + whisper.cpp EN via sherpa-onnx	§7 · CER table (P0)
STT forms sentences on pauses, streams instantly	Silero VAD sentence chunking inside PTT hold; per-sentence frames transmitted as formed	§5 stage 2 · live demo
Stream via Wi-Fi/Bluetooth to embedded device or phone	NSD+TCP · BT RFCOMM · ESP32 LoRa bridge — one frame format, three bearers	§5, §6.4 · demo legs
TTS converts received text to intelligible speech, voice-note style	FastPitch+HiFi-GAN / Piper, sentence-streamed, normalized text, pronunciation lexicon	§7 · listening demo
Alert messages: highest volume, non-interruptible	ALERT-priority frames → audio-focus-exclusive max-volume playback, repeat policy	§6.5 · alert demo
Two-phone loop: STT mode + TTS mode = walkie-talkie (PTT); off = normal phone	Exactly this, as the flagship demo, plus latency timer on screen	§8 loop verification
Open-source only; TFLite/PyTorch Mobile-class frameworks; no internet APIs	MIT/Apache/GPL-clean tree; ONNX Runtime (sherpa-onnx); zero network calls — license audit table in repo	§7.3 box, §9
Metrics: efficiency/accuracy/latency evidence	Live Evaluation Mode + reproducible desktop harness (CSV + scripts in repo)	§6.3, §8

Submission artifacts: idea PPT (6 slides, template-compliant) · this idea document · GitHub repo (app + firmware + eval harness + license audit) · demo video of the two-phone loop · benchmark CSVs.

§14 Research & References

Grouped by section. All links verified during research (Aug 2026). Full research dossiers (5 documents, ~1,000 lines) live in the repo under [research/](#).

Problem & India context (§2)

1. [R1] Amphan: networks at 65–70% capacity (COAI) — [Deccan Herald](#)
2. [R2] Amphan: ~7,000/14,167 WB towers down — [The Quint](#); WMO evacuation praise — [WMO](#)
3. [R3] Himachal floods 2025: ~66% of Chamba telecom sites down — [Diary Times](#)
4. [R4] ~770,000 towers disaster-exposed (CDRI) — [Business Standard](#)
5. [R5] DoT+CDRI Disaster Risk & Resilience Framework — [PIB](#)
6. [R6] Cyclone Fani comms lessons — [The Conversation](#)
7. [R7] Literacy in India (rural/urban, gender) — [Census/NSO compilation](#)
8. [R8] State literacy rates — [NSO/survey compilation](#)
9. [R9] Rural mother-tongue-only communication (~90%) — [Kadence Digital India study](#)
10. [R10] SACHET scale & cell broadcast — [NDMA](#), [PIB](#)
11. [R11] AIR/DD reinforced for last-mile alerts — [News on Air](#)
12. [R12] IVR design for low-literacy users — [ICTD literature](#)
13. [R13] Speech interfaces for low-literate users — [ACM DEV](#)

Bitrate science & semantic communication (§4)

14. [R14] Codec2 (450–3200 bps) — [GitHub](#); FreeDV — [freedv.org](#)
15. [R15] MELPe / STANAG 4591 (600/1200/2400 bps) — [melp.org](#)
16. [R16] Google Lyra v2 — [Google Open Source blog](#)
17. [R17] Meta EnCodec — [arXiv 2210.13438](#)
18. [R18] Speaking rate 130–150 wpm — [VirtualSpeech](#)
19. [R19] ISRO Disaster Management Support programme — [overview](#); NESAC SATCOM — [nesac.gov.in](#)
20. [R20] Ham radio in disaster comms — [reference](#)
21. [R29] Qin, Tao, Lu, Tong, Li — Semantic Communications: Principles and Challenges — [arXiv 2201.01389](#)
22. [R30] DeepSC-ST: semantic comms with speech recognition & synthesis — [IEEE TWC 2023](#); DeepSC-S — [arXiv 2102.12605](#)
23. [R31] STCTS ~80 bps semantic speech codec — [arXiv 2512.00451](#)
24. [R32] Voice over Low Data Rate Networks Using STT & Semantic Compression (94% latency cut) — [2025 paper](#)
25. [R33] FEMA IPAWS: TTS mispronunciation guidance — [FEMA Tip 42](#)

Prior art & engineering (§3, §5, §9)

34. [R34] Oboe low-latency audio — [google/oboe](#)
35. [R35] Silero VAD — [GitHub](#); VAD comparison — [Picovoice](#)
36. [R36] IndicConformer (MIT, 22 languages, int8 ONNX) — [HF model card](#)
37. [R37] sherpa-onnx — [k2-fsa/sherpa-onnx](#); IndicConformer conversion thread — [discussion #3199](#)
38. [R38] Vistaar/IndicWhisper benchmarks — [arXiv 2305.15386](#); whisper.cpp — [GitHub](#)
39. [R39] indic-numtowords — [AI4Bharat](#); Indic NLP Library — [docs](#)
40. [R40] LANwalkieTalkie (socket PTT skeleton) — [GitHub](#); bitchat-android BLE mesh — [GitHub](#); Bridgefy protocol break — [IACR 2021/214](#)
41. [R41] AI4Bharat Indic-TTS (MIT, FastPitch+HiFi-GAN, 13 languages) — [GitHub](#); paper — [arXiv 2211.09536](#)
42. [R42] Piper TTS voices — [manifest](#); ARM RTF — [RK3588 benchmark](#)
43. [R43] Streaming TTS first-audio latency — [Soniox](#)
44. [R44] WER wrong for agglutinative Indic; use CER — [arXiv 2203.16601](#); SN-WER — [arXiv 2606.02548](#)
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46. [R46] Meshtastic LoRa range/bitrate — [DIY guide](#)

- 47. [R47] ESP32+SX1278 India pricing — [Compo India](#)
 - 48. [R48] IN865 band plan, 1% duty, 30 dBm — [Ensemble Tech](#); QMesh — [Hackaday](#); codec2_talkie — [GitHub](#)
 - 49. [R49] J-Alert TTS multilingual alerts — [reference](#)
 - 50. [R50] Bhashini / NLTM (₹450 crore) — [IndiaAI](#)
 - 51. [R51] RTranslator (offline STT→MT→TTS, Apache-2.0) — [GitHub](#)
 - 52. [R52] MMS-TTS CC-BY-NC license — [fairseq MMS README](#)
 - 53. [R53] int8 quantization $\approx$ free for ASR — [arXiv 2511.08093](#), [arXiv 2408.03979](#)
- Repo: private GitHub [itantra](#) — research dossiers, this document, deck, eval harness, and code, committed continuously.